

SIMS Working Group Accomplishments Report

JETSCAPE 2025 Collaboration Meeting

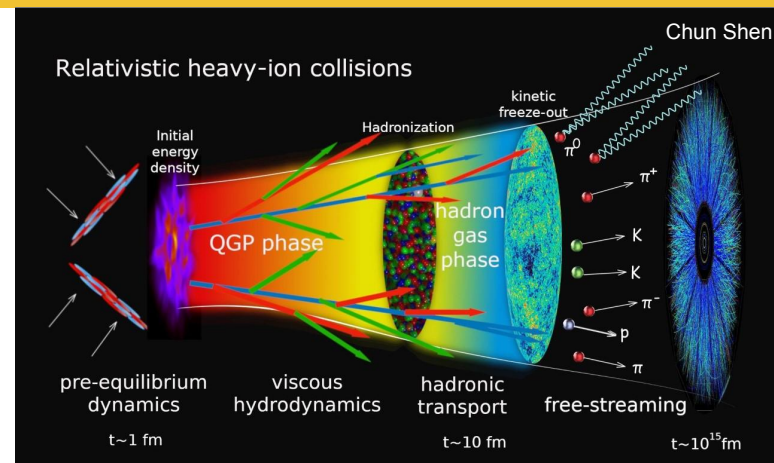
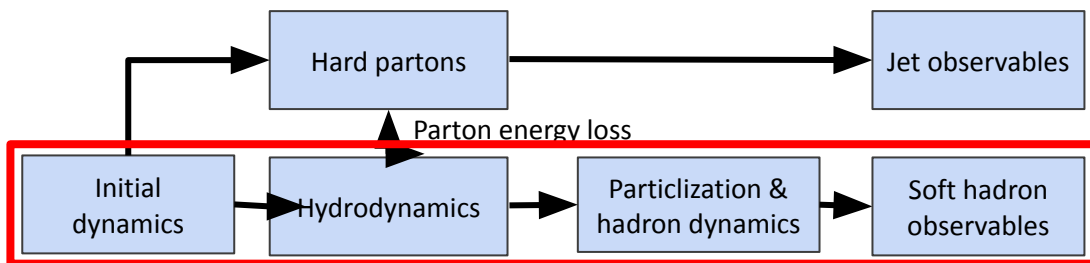
Andi Mankolli on behalf of the SIMS Working Group
December 17, 2025



X-SCAPE

Overview - SIMS Working Group

- **SIM**ulation**S** and distributed computing (SIMS) working group
 - Soft sector of heavy ion collisions



Studying “the most perfect fluid”

Overview - SIMS Accomplishments

- 3D calibration of soft sector
 - Currently undergoing part II of Collaboration Review

Longitudinal dynamics of large and small systems from a 3D Bayesian calibration of
RHIC top-energy collision data

A. Mankolli,¹ C. Shen,^{2,3} M. Luzum,⁴ J.-F. Paquet,¹ M. Singh,¹ J. Velkovska,¹ S. A. Bass,⁵ C. Gale,⁶
G. A. C. da Silva,⁴ L. Du,^{7,8,6} L. Kasper,¹ G. S. Rocha,^{9,1} D. Soeder,⁵ S. Tuo,¹ G. Vujanovic,¹⁰
X. Wu,⁶ W. Zhao,^{7,8} M. Chartier,¹¹ Y. Chen,¹ R. Datta,² R. Dolan,² R. Ehlers,^{7,8} H. Elfner,^{12,13,14}
R. J. Fries,^{15,16} D. A. Hangal,^{17,2} B. V. Jacak,^{7,8} P. M. Jacobs,^{7,8} S. Jeon,⁶ Y. Ji,¹⁸ F. Jonas,^{7,8}
M. Kordell II,^{15,16} A. Kumar,¹⁰ R. Kunnawalkam-Elayavalli,¹ J. Latessa,¹⁹ Y.-J. Lee,^{20,21} A. Majumder,²
S. Mak,¹⁸ C. Martin,²² H. Mehryar,¹⁹ T. Mengel,²² C. Nattrass,²² J. Norman,²³ M. Ockleton,¹¹
C. Parker,^{15,16} J. H. Putschke,² H. Roch,² G. Roland,^{20,21} B. Schenke,²⁴ L. Schwiebert,¹⁹ A. Sengupta,^{15,16}
C. Sirimanna,⁵ R. A. Soltz,^{17,2} I. Soudi,^{25,26} Y. Tachibana,²⁷ X.-N. Wang,^{28,7,8} and J. Zhang¹
(The JETSCAPE Collaboration)

- Predictions using SIMS tune for new and upcoming measurements

Overview - SIMS Accomplishments

- 3D calibration of soft sector
 - Currently undergoing part II of Collaboration Review

Longitudinal dynamics of large and small systems from a 3D Bayesian calibration of RHIC top-energy collision data

A. Mankolli,¹ C. Shen,^{2,3} M. Luzum,⁴ J.-F. Paquet,¹ M. Singh,¹ J. Velkovska,¹ S. A. Bass,⁵ C. Gale,⁶
G. A. C. da Silva,⁴ L. Du,^{7,8,6} L. Kasper,¹ G. S. Rocha,^{9,1} D. Soeder,⁵ S. Tuo,¹ G. Vujanovic,¹⁰
X. Wu,⁶ W. Zhao,^{7,8} M. Chartier,¹¹ Y. Chen,¹ R. Datta,² R. Dolan,² R. Ehlers,^{7,8} H. Elfner,^{12,13,14}
R. J. Fries,^{15,16} D. A. Hangal,^{17,2} B. V. Jacak,^{7,8} P. M. Jacobs,^{7,8} S. Jeon,⁶ Y. Ji,¹⁸ F. Jonas,^{7,8}
M. Kordell II,^{15,16} A. Kumar,¹⁰ R. Kunnawalkam-Elayavalli,¹ J. Latessa,¹⁹ Y.-J. Lee,^{20,21} A. Majumder,²
S. Mak,¹⁸ C. Martin,²² H. Mehryar,¹⁹ T. Mengel,²² C. Nattrass,²² J. Norman,²³ M. Ockleton,¹¹
C. Parker,^{15,16} J. H. Putschke,² H. Roch,² G. Roland,^{20,21} B. Schenke,²⁴ L. Schwiebert,¹⁹ A. Sengupta,^{15,16}
C. Sirimanna,⁵ R. A. Soltz,^{17,2} I. Soudi,^{25,26} Y. Tachibana,²⁷ X.-N. Wang,^{28,7,8} and J. Zhang¹
(The JETSCAPE Collaboration)

- Predictions using SIMS tune for new and upcoming measurements

Overview: 3D Calibration with RHIC Data

- 2D JETSCAPE calibration of the soft sector:
 - Phys. Rev. C 103, 054904 [Cited by 399](#)
 - Phys. Rev. Lett. 126, 242301 [Cited by 238](#)
- 3D calibration
 - 3D Glauber initial condition (**Trento 3D in future**)*
 - RHIC 200 GeV data (**LHC in future**)*
 - Au-Au and d-Au (plus additional small systems)
- Since last meeting
 - Finalized the calibration dataset
 - Extensive investigations of longitudinal effects with calibrated model
 - Tension investigations and additional cross-checks
 - LHC adapted tune
 - Writing of manuscript

*talk by J-F Paquet

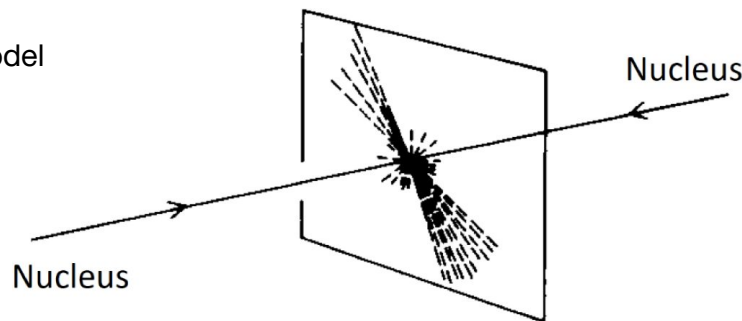
Multisystem Bayesian constraints on the transport coefficients of QCD matter

[D. Everett](#)¹, [W. Ke](#)^{2,3}, [J.-F. Paquet](#)⁴, [G. Vujanovic](#)⁵, [S. A. Bass](#)⁴, [L. Du](#)¹, [C. Gale](#)⁶, [M. Heffernan](#)⁶, [U. Heinz](#)¹ *et al.* (JETSCAPE Collaboration)

Longitudinal dynamics of large and small systems from a 3D Bayesian calibration of RHIC top-energy collision data

A. Mankolli,¹ C. Shen,^{2,3} M. Luzum,⁴ J.-F. Paquet,¹ M. Singh,¹ J. Velkovska,¹ S. A. Bass,⁵ C. Gale,⁶ G. A. C. da Silva,⁴ L. Du,^{7,8,6} L. Kasper,¹ G. S. Rocha,^{9,1} D. Soeder,⁵ S. Tuo,¹ G. Vujanovic,¹⁰ X. Wu,⁶ W. Zhao,^{7,8} M. Chartier,¹¹ Y. Chen,¹ R. Datta,² R. Dolan,² R. Ehlers,^{7,8} H. Elfner,^{12,13,14} R. J. Fries,^{15,16} D. A. Hangal,^{17,2} B. V. Jacak,^{7,8} P. M. Jacobs,^{7,8} S. Jeon,⁶ Y. Ji,¹⁸ F. Jonas,^{7,8} M. Kordell II,^{15,16} A. Kumar,¹⁰ R. Kunnawalkam-Elayavalli,¹ J. Latessa,¹⁹ Y.-J. Lee,^{20,21} A. Majumder,² S. Mak,¹⁸ C. Martin,²² H. Mehryar,¹⁹ T. Mengel,²² C. Nattrass,²² J. Norman,²³ M. Ockleton,¹¹ C. Parker,^{15,16} J. H. Putschke,² H. Roch,² G. Roland,^{20,21} B. Schenke,²⁴ L. Schwiebert,¹⁹ A. Sengupta,^{15,16} C. Sirimanna,⁵ R. A. Soltz,^{17,2} I. Soudi,^{25,26} Y. Tachibana,²⁷ X.-N. Wang,^{28,7,8} and J. Zhang¹

(The JETSCAPE Collaboration)



The model: 3D Initial State and Hydrodynamics

Impose energy
momentum
conservation

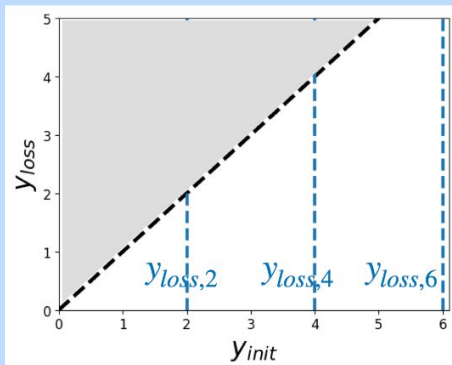
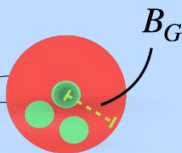
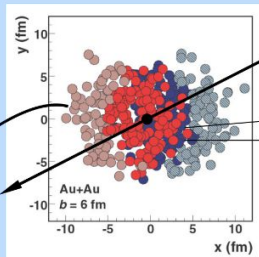
α_{rem}
(Remnant
energy loss
fraction)

Schenke, Shen

Phys. Rev. C 105, 064905
(2022)

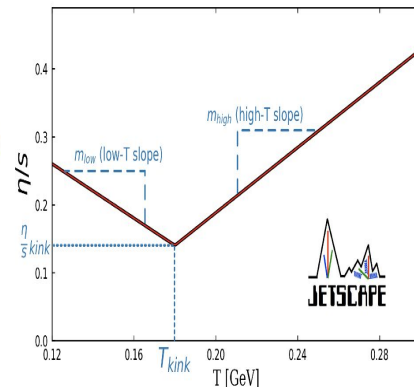
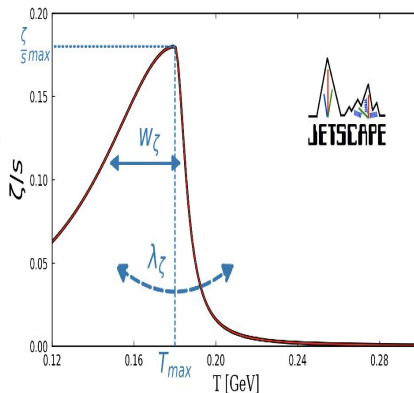
Phys. Rev. C 97, 024907
(2018)

Initial State



3D MCGlauber

Hydrodynamics



Hadronic Transport

MUSIC + UrQMD

Model run through iEBE framework. We checked that simulations can be reproduced in X-SCAPE

Experimental Data



Au-Au 200 GeV

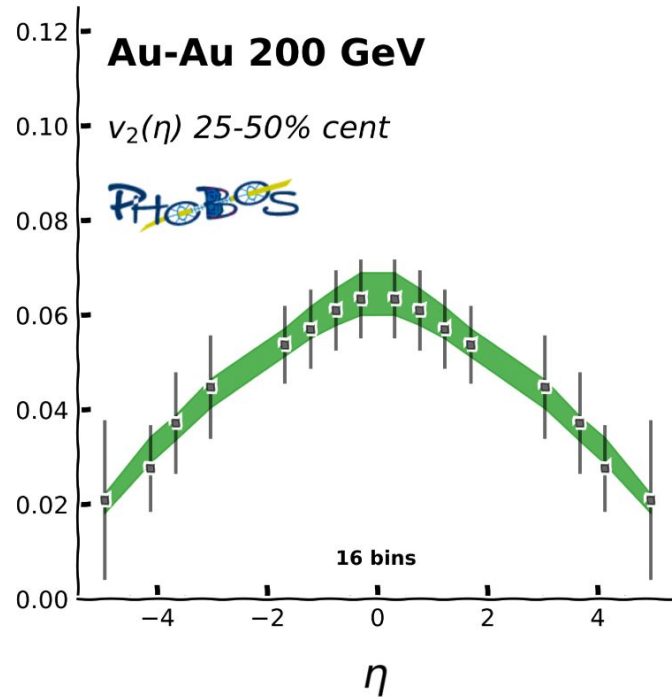
- $v_2(\eta)$ STAR
- $v_2(\eta)$ PHOBOS
- $\langle p_T \rangle \pi, k$ STAR
- $\langle p_T \rangle k, p$ PHENIX
- $\langle p_T \rangle p$ STAR
- $\langle p_T \rangle \pi$ PHENIX
- $dN_{ch}/d\eta(\eta)$ PHOBOS
- $v_2(\text{cent})$ STAR
- $v_3(\text{cent})$ STAR
- $v_2(p_T)$ PHENIX
- $v_2(p_T)$ STAR
- $v_3(p_T)$ PHENIX
- $v_4(p_T)$ PHENIX
- $dN_{ch}/d\eta(\eta)$ BRAHMS
- $r_2(\eta)$ STAR
- $r_3(\eta)$ STAR
- $dE_T/d\eta(\text{cent})$ PHENIX

d-Au 200 GeV

- $dN_{ch}/d\eta(\eta)$ PHOBOS
- $dN_{ch}/d\eta(\eta)$ PHENIX
- $v_2(p_T)$ PHENIX
- $v_2(p_T)$ STAR
- $v_2(\eta)$ PHENIX
- $v_3(p_T)$ STAR
- $v_3(p_T)$ PHENIX
- $dE_T/d\eta(\text{cent})$ PHENIX

Experimental Data - Modeling the Observables

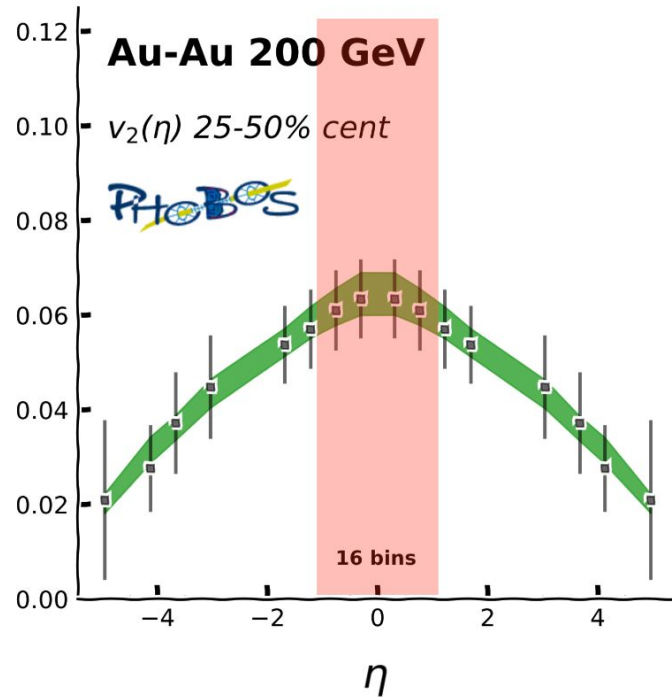
Example:



- “Midrapidity” measurements contain forward/backward information
- Centrality selection
 - $3.5 < |\eta| < 4.5$
- Event plane determination
 - $2.05 < |\eta| < 3.2$

Experimental Data - Modeling the Observables

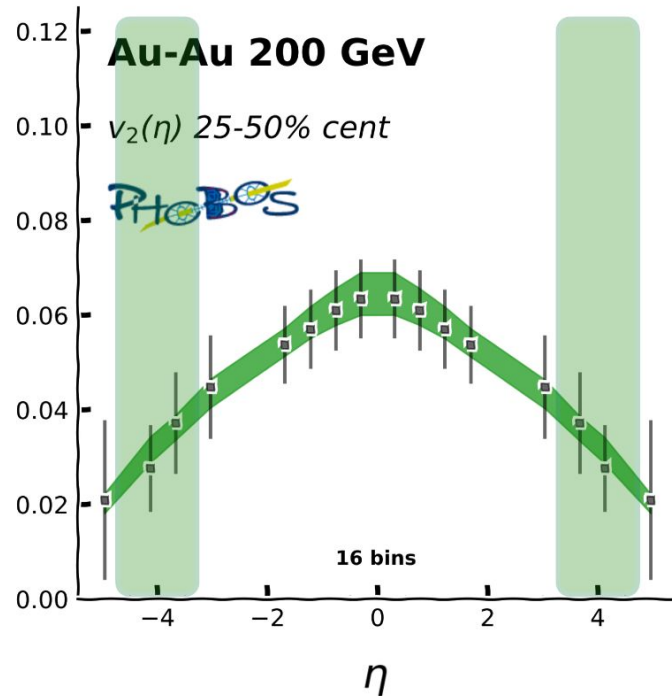
Example:



- “Midrapidity” measurements contain forward/backward information
- Centrality selection
 - ▶ $3.5 < |\eta| < 4.5$
- Event plane determination
 - ▶ $2.05 < |\eta| < 3.2$

Experimental Data - Modeling the Observables

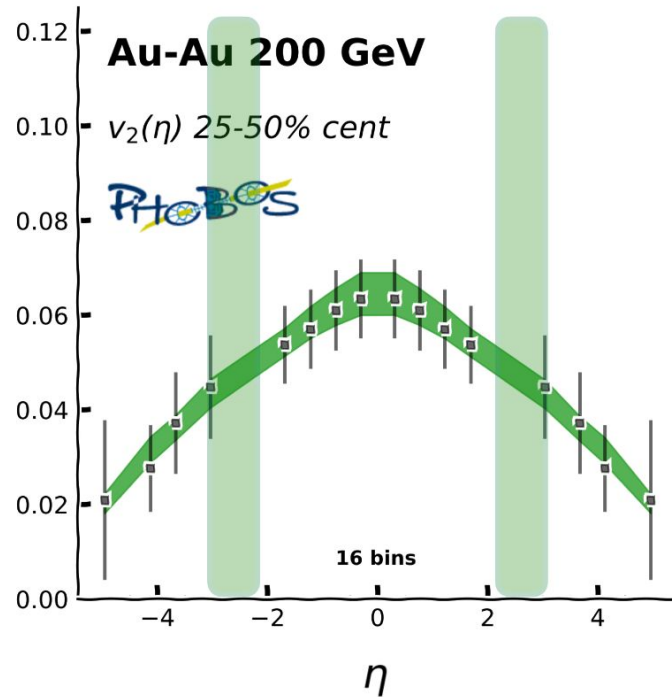
Example:



- “Midrapidity” measurements contain forward/backward information
- Centrality selection
 - ▶ $3.5 < |\eta| < 4.5$
- Event plane determination
 - ▶ $2.05 < |\eta| < 3.2$

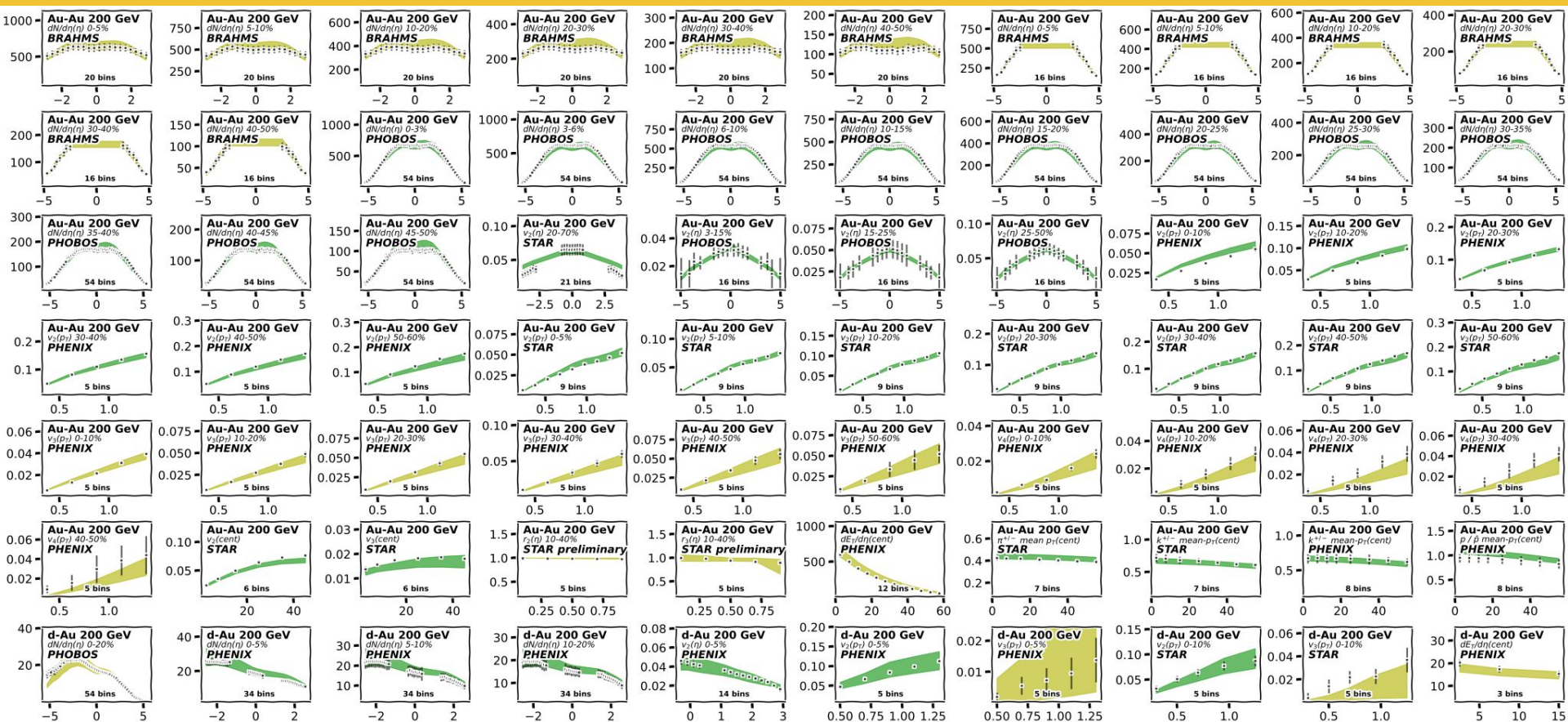
Experimental Data - Modeling the Observables

Example:

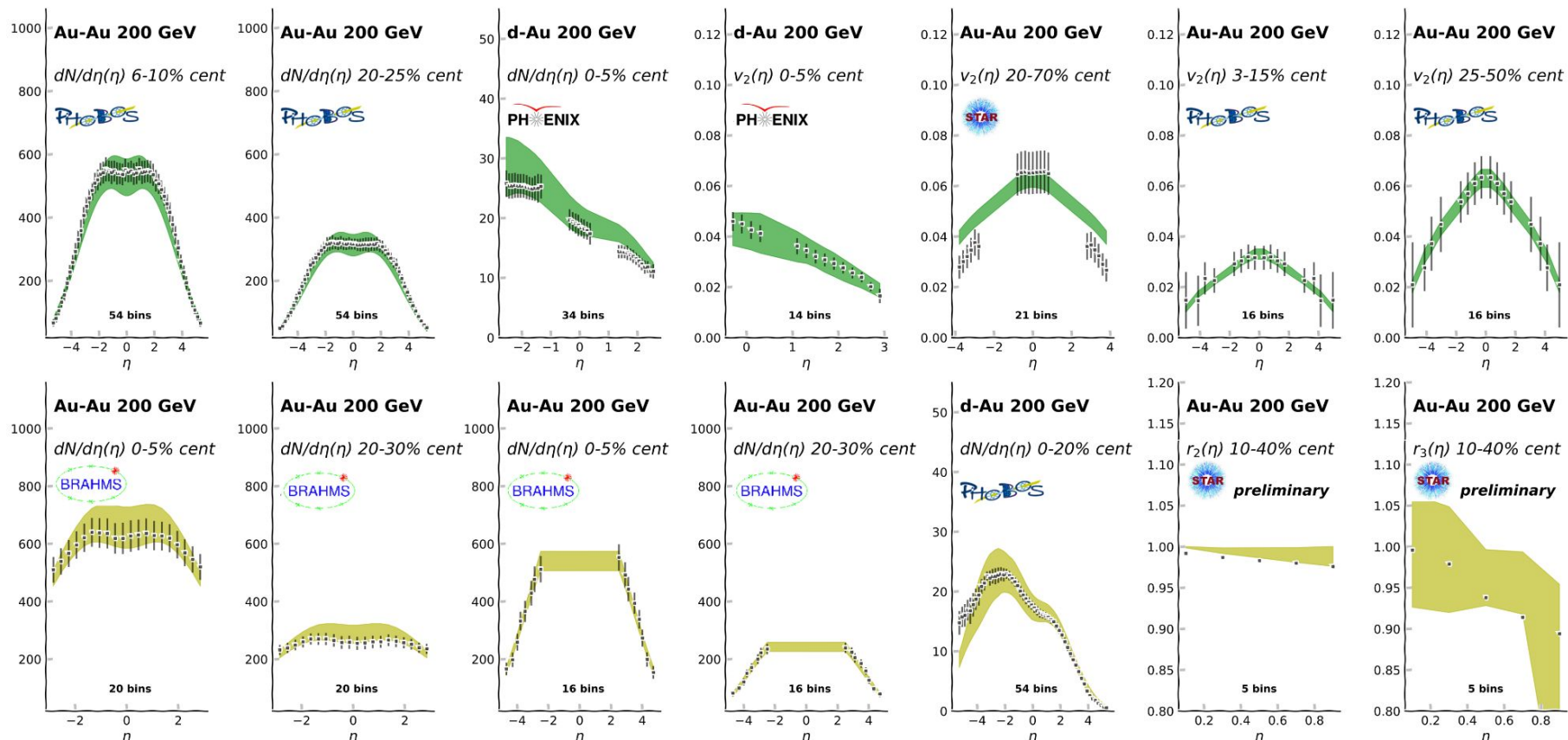


- “Midrapidity” measurements contain forward/backward information
- Centrality selection
 - ▶ $3.5 < |\eta| < 4.5$
- Event plane determination
 - ▶ $2.05 < |\eta| < 3.2$

Observables from Calibrated Model

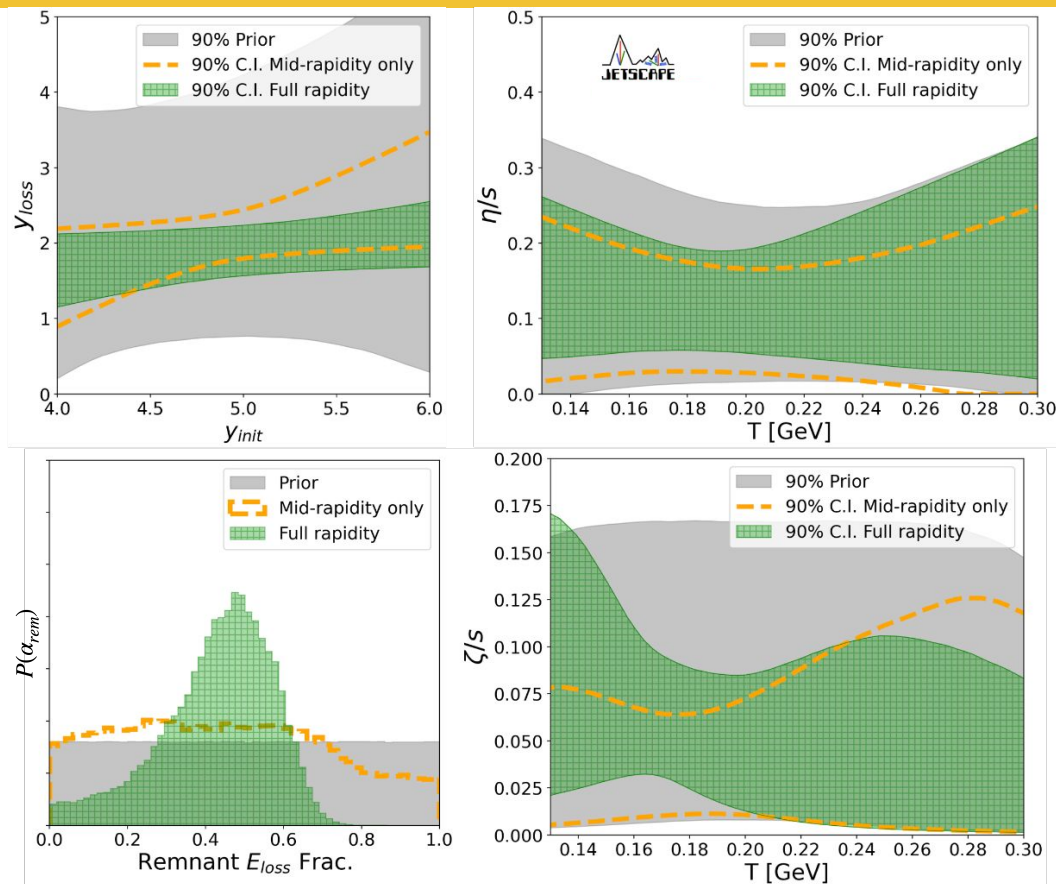


Rapidity-dependent Observable Posterior



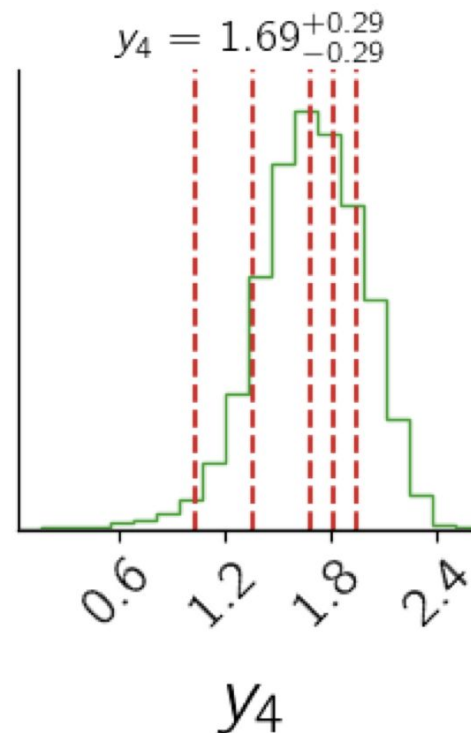
Constraints from Forward/Backward Data

- Significant rapidity loss constraints provided by mid-rapidity data
- E_{loss} of nuclear fragments constrained by finite rapidity data
- Viscosity posteriors shifted to larger values
 - Preference for finite bulk viscosity at low T



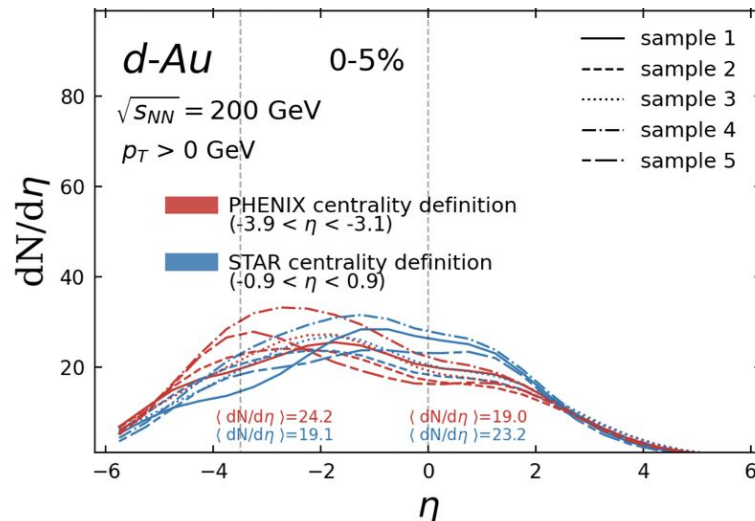
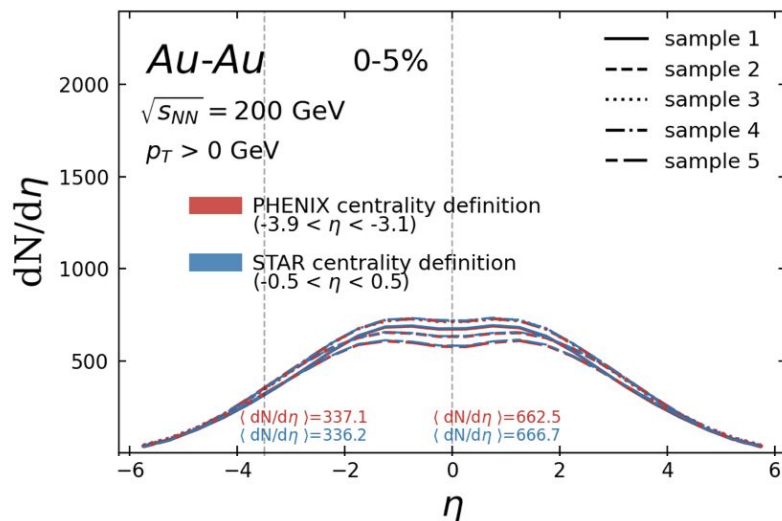
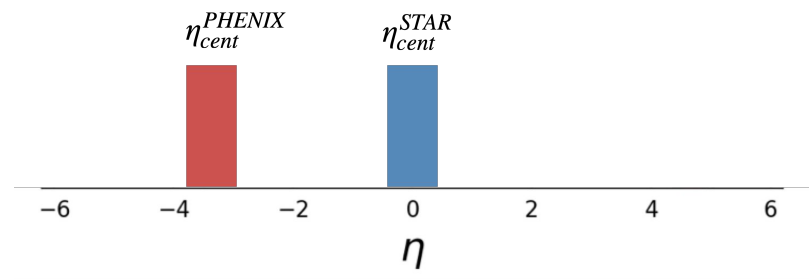
Predictions with the Calibrated Model

- Can we use the **calibrated model** to quantify the impact of generic experimental choices on observables?
- Two test cases:
 - **Centrality selection** for $dN/d\eta$
 - **Reference region** for $v_n(\eta)$
- Calculations using **5 high-statistics samples** as a representation of the posterior



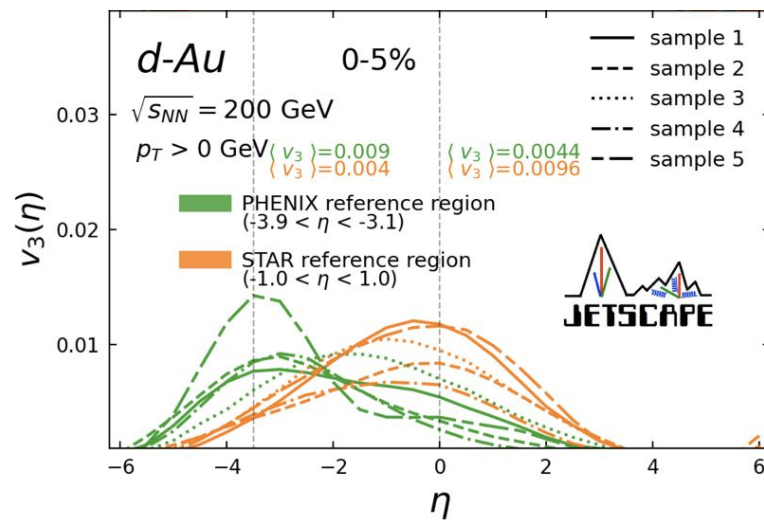
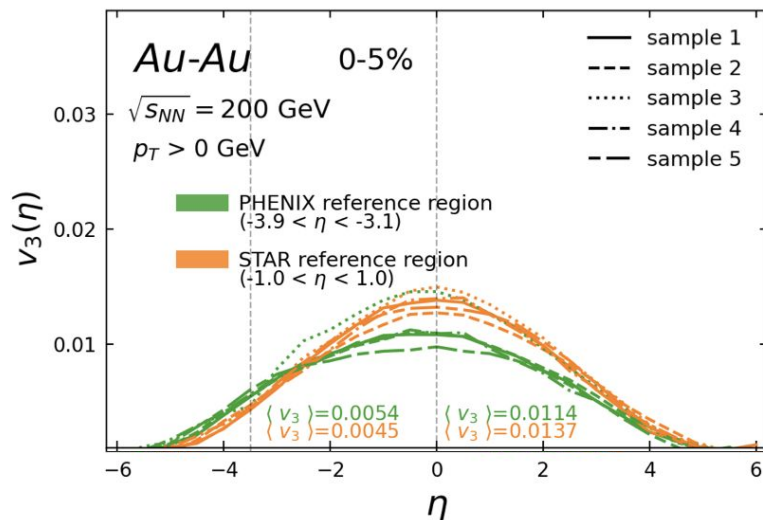
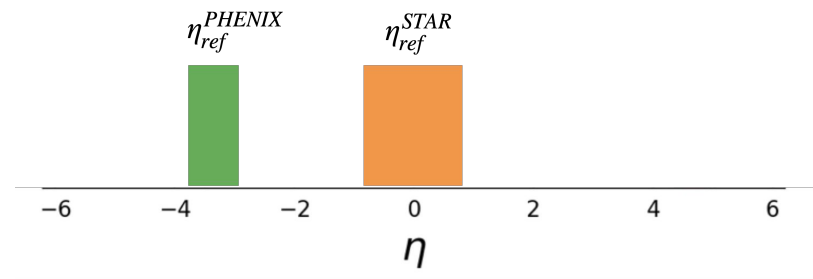
Calculations with the Calibrated Model: Centrality Selection

- The effect of centrality selection region on $dN/d\eta$ is small ($<1\%$) in Au-Au but large ($\sim 20\%$) in d-Au



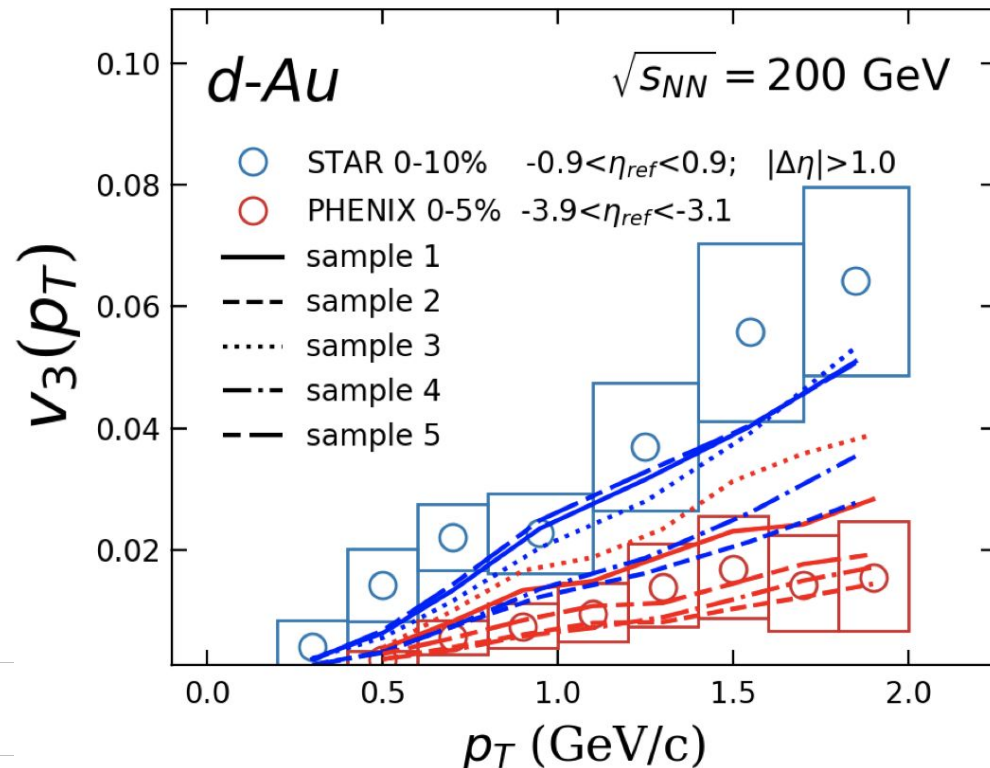
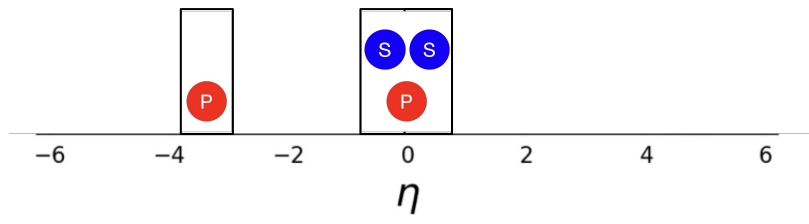
Calculations with the Calibrated Model: Reference region

- The effect of selection of the rapidity reference region on $v_3(\eta)$ is **~15-20%** in Au-Au and **~50%** in d-Au



Testing the Calibrated Model: STAR vs PHENIX $v_3(p_T)$ in d-Au

- The calibrated model can account for **some of the difference** between STAR and PHENIX $v_3(p_T)$ measurements in d-Au

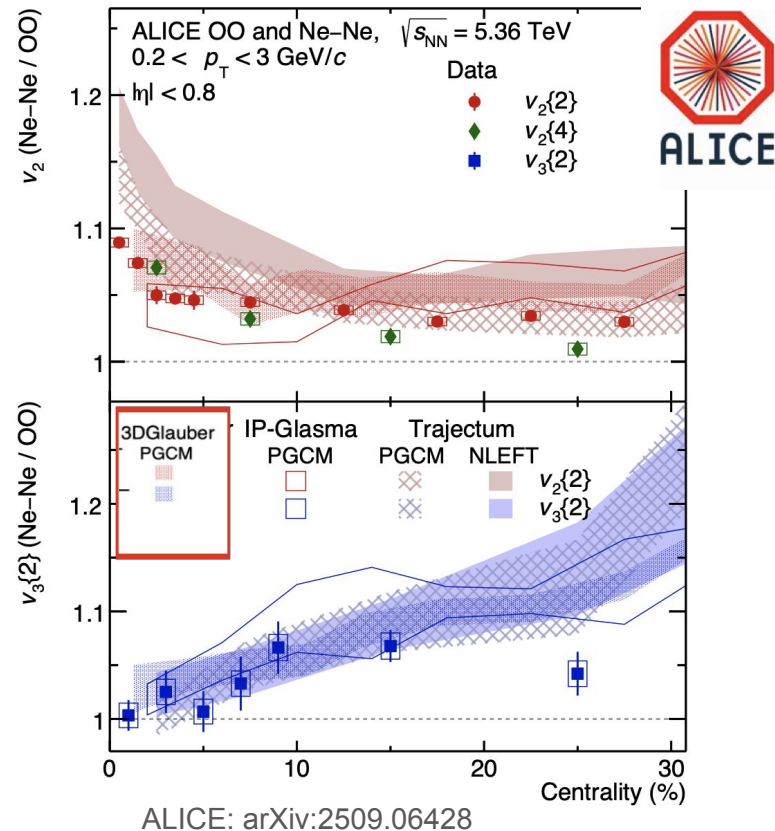


Overview - SIMS Accomplishments

- 3D calibration of soft sector
 - Currently undergoing part II of Collaboration Review
- Predictions using SIMS tune for new and upcoming measurements

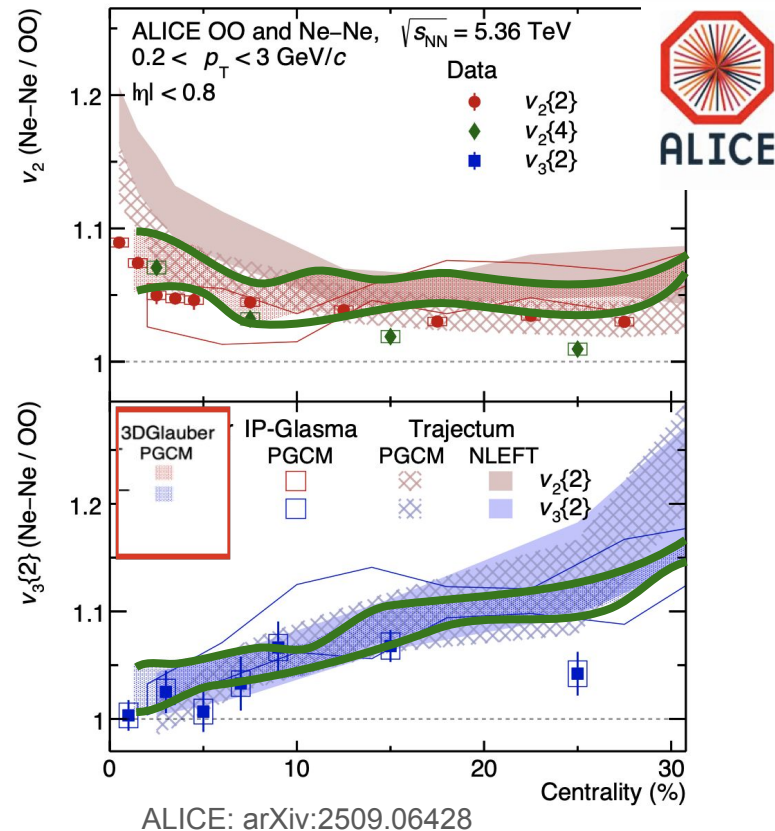
Predictions for New Measurements: LHC tune

- Starting from the MAP, we obtained an LHC tune by fitting one more parameter (y_{10})
- Used LHC tune to make light ion (O-O, Ne-Ne) predictions for v_n
 - Mid-rapidity measurement in symmetric collision systems
- Shown in ALICE talk at Initial Stages Conference in Sep. 2025
- Working on forward/backward mean p_T in p-Pb requested by LHCb



Predictions for New Measurements: LHC tune

- Starting from the MAP, we obtained an LHC tune by fitting one more parameter (y_{10})
- Used LHC tune to make light ion (O-O, Ne-Ne) predictions for v_n
 - Mid-rapidity measurement in symmetric collision systems
- Shown in ALICE talk at Initial Stages Conference in Sep. 2025
- Working on forward/backward mean p_T in p-Pb requested by LHCb



Summary - SIMS Accomplishments

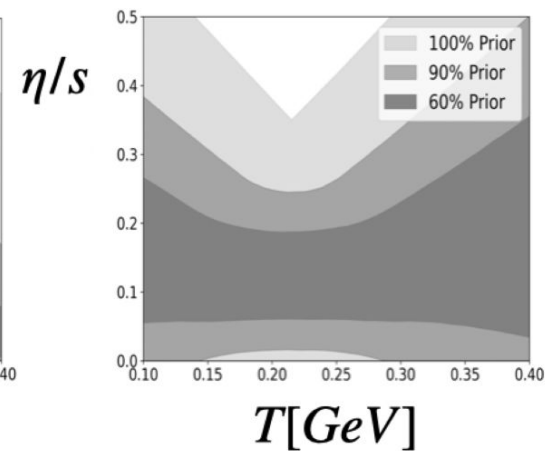
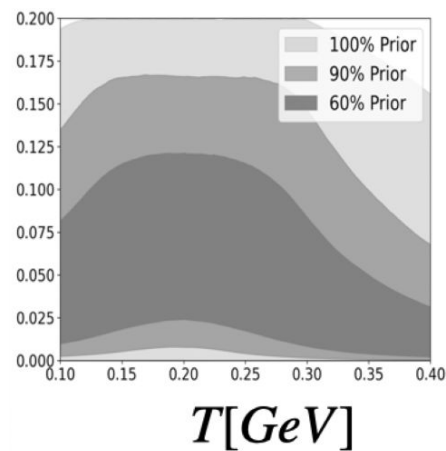
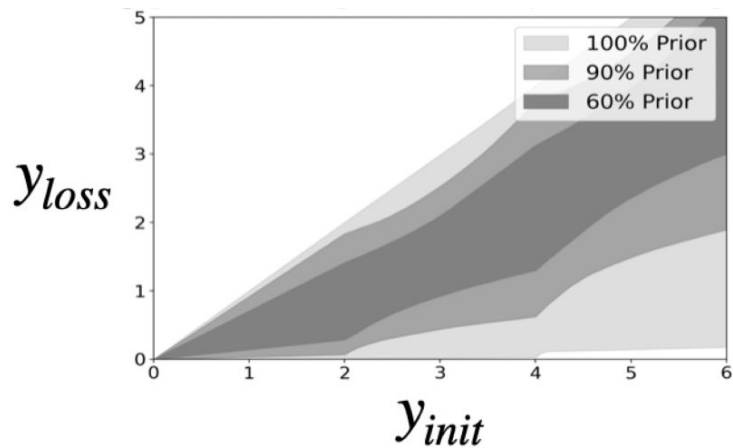
- 3D calibration of soft sector
 - Many new results on the longitudinal structure of heavy ion collisions and its impact on model constraints
 - Finished writing the manuscript
 - Currently undergoing part II of Collaboration Review
 - Already used as a tune for future studies in JETSCAPE and to make predictions for new measurements (v_n in O-O, Ne-Ne)

Backup

The Model Parameters & Priors

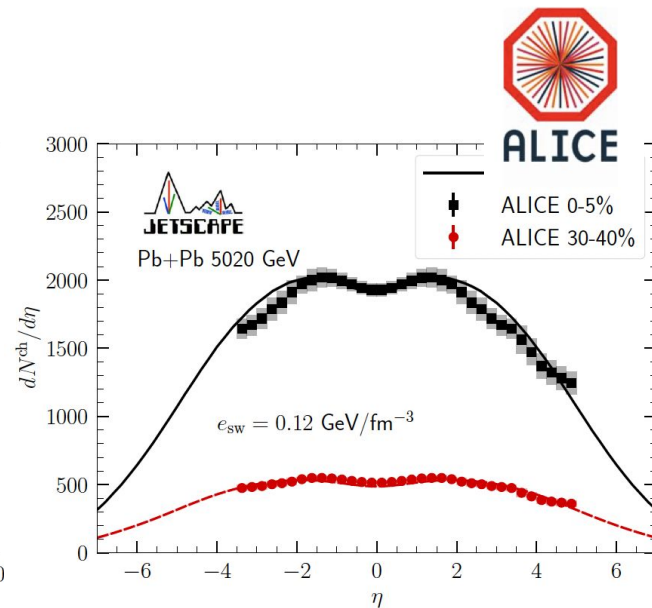
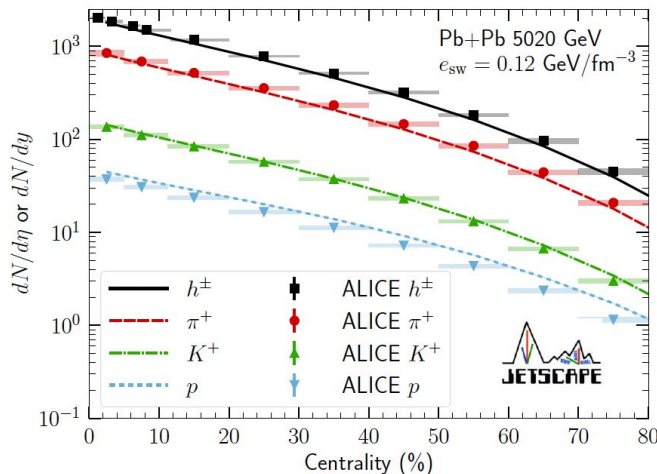
Parameter	Collision Stage	Prior Range
$y_{loss,2}$	Initial State	[0,2]
$y_{loss,4}$	Initial State	$[y_{loss,2},4]$
$y_{loss,6}$	Initial State	$[y_{loss,4},6]$
$\sigma_{y_{loss}}$	Initial State	[0,1]
α_{rem}	Initial State	[0,1]
Shadowing Factor	Initial State	[0,1]
τ_{form} Mean	Initial State	[0.2,1]
B_G [1/GeV ²]	Initial State	[2,25]
String Source σ_x [fm]	Initial State	[0.1,0.5]
String Source σ_η	Initial State	[0.1,0.8]
String Trans. Shift Frac.	Initial State	[0,1]
$\frac{\eta}{s} T_{kink}$ [GeV]	Hydro	[0.13,0.3]
$\frac{\eta}{s}$ low-T slope	Hydro	[-2,1]
$\frac{\eta}{s}$ high-T slope	Hydro	[-1,2]
$\frac{\eta}{s}$ at kink	Hydro	[0.01,0.2]
ζ_s max	Hydro	[0.01,0.2]
$\zeta_s T_{peak}$ [GeV]	Hydro	[0.12,0.3]
ζ_s width	Hydro	[0.025,0.15]
$\zeta_s \lambda$ assym.	Hydro	[-0.8,0.6]
EPS Switch [GeV/fm ³]	Particlization	[0.1,0.6]

Plots of Parameter Priors



LHC Tune: Comparison with Pb-Pb Data

- Soft sector LHC tune to study soft-hard correlations in PHYS
 - See talk by Chun on Mon. 12/15
- Manually tuned rapidity loss for large incoming rapidity (y_{10})
- Tuned e_{switch} at LHC to fit protons



Tension investigations

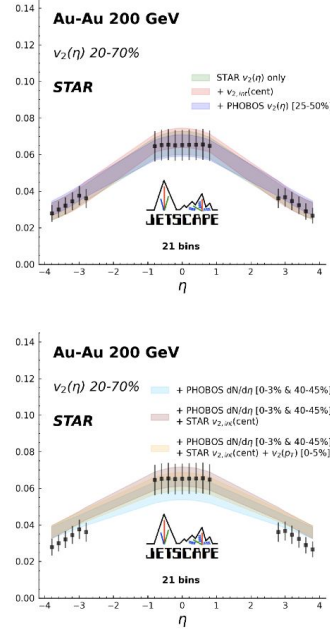


Figure 9. The comparison of the STAR $v_2(\eta)$ posterior with the measurement, showing calibrations with various subsets of the data and highlighting the model tension between certain observables and the fit to the STAR measurement. The top plot shows calibrations using subsets of v_2 data only. The bottom plot shows calibrations that include subsets of multiplicity data. All data in these calibrations are from Au-Au measurements.

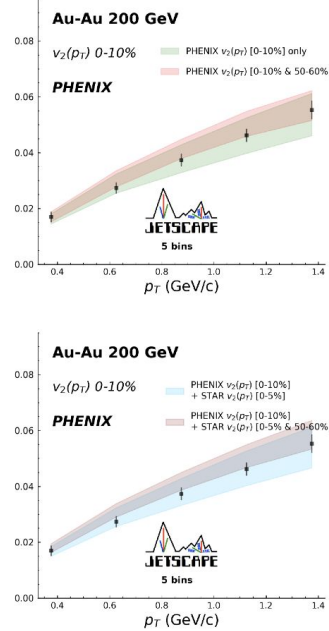
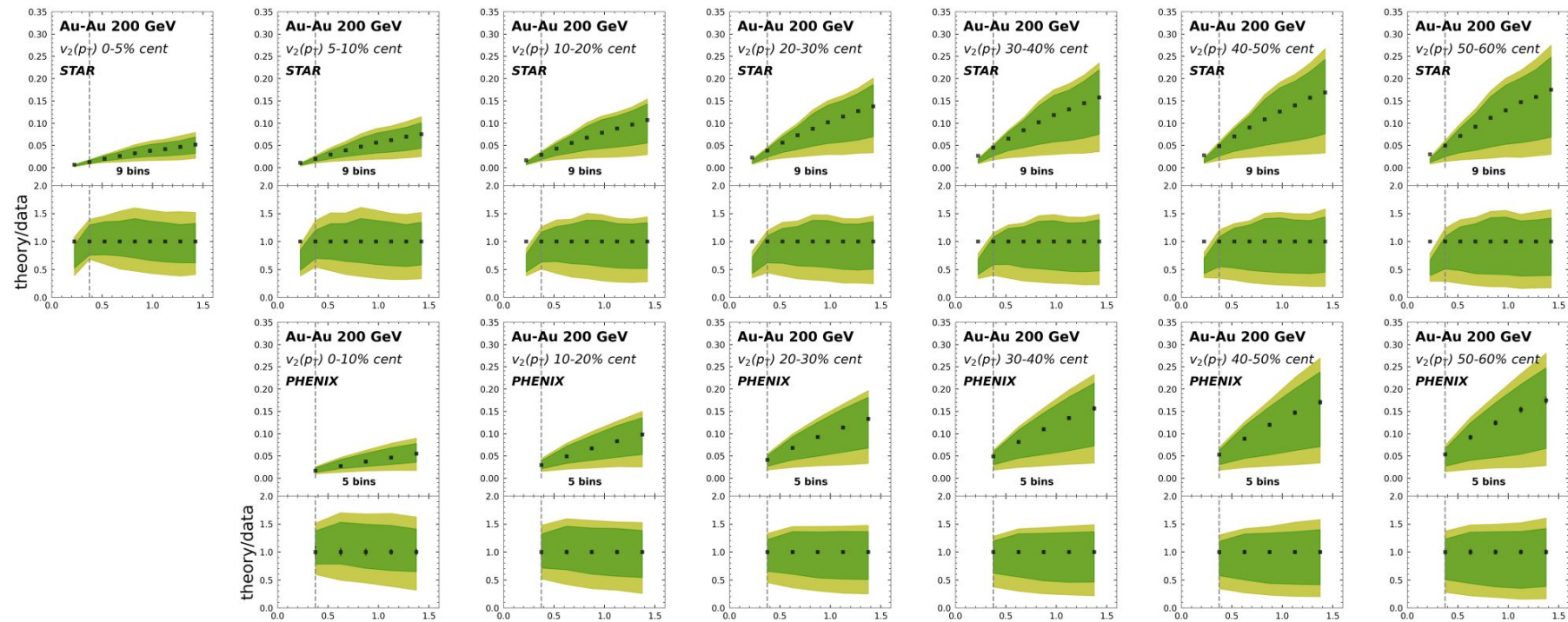
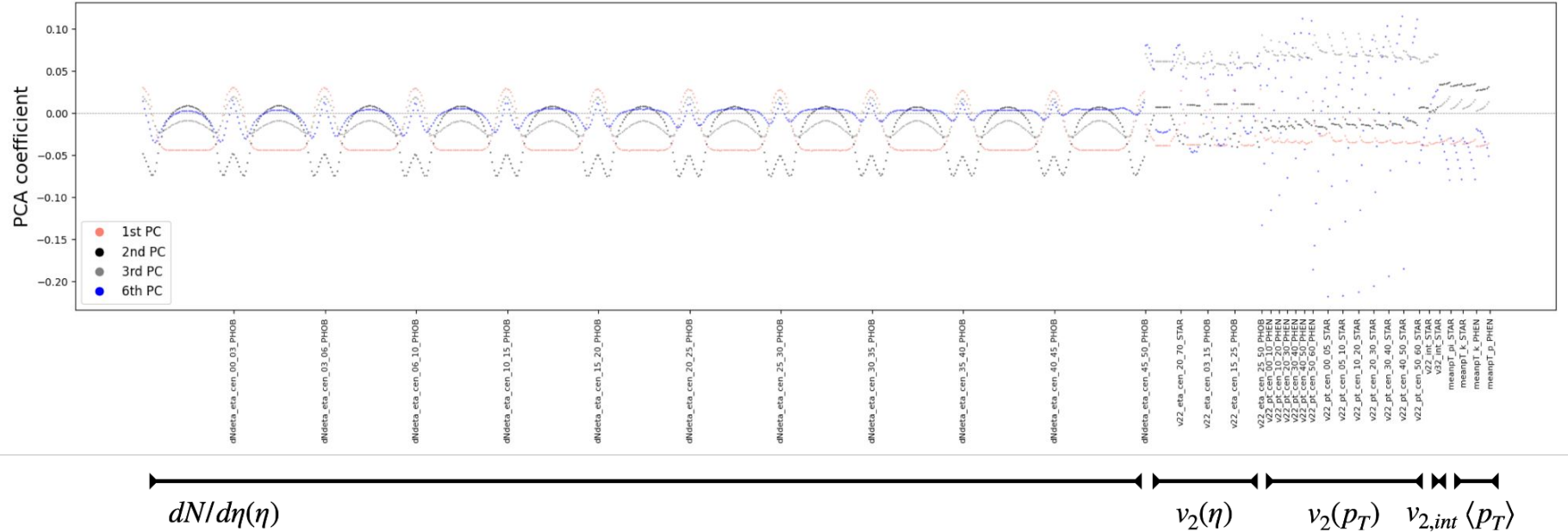


Figure 10. The comparison of the PHENIX $v_2(p_T)$ posterior for central (0-10%) Au-Au collisions with the measurement, showing calibrations with various subsets of the data and highlighting the model tension between different bins of v_2 data and the fit to the PHENIX measurement. The top plot shows calibrations using subsets of PHENIX v_2 data only. The bottom plot shows calibrations that include subsets of STAR data. All data in these calibrations are from Au-Au measurements.

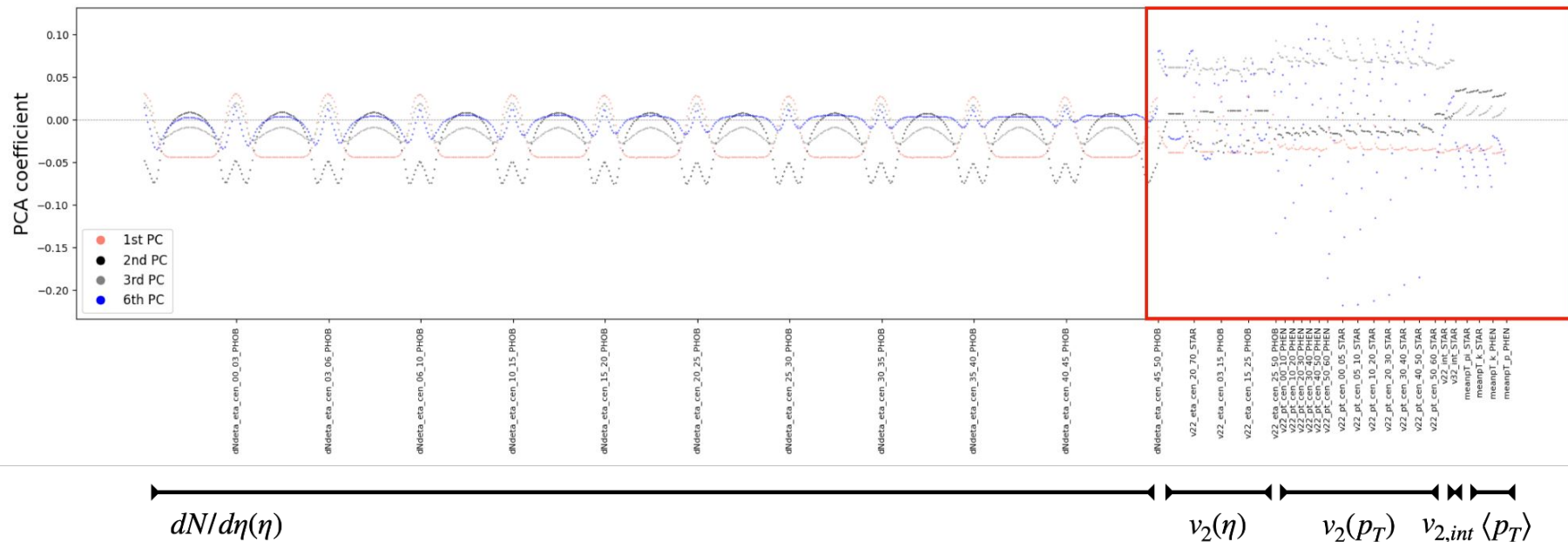
Revisiting $v_2(p_T)$ - observable posterior



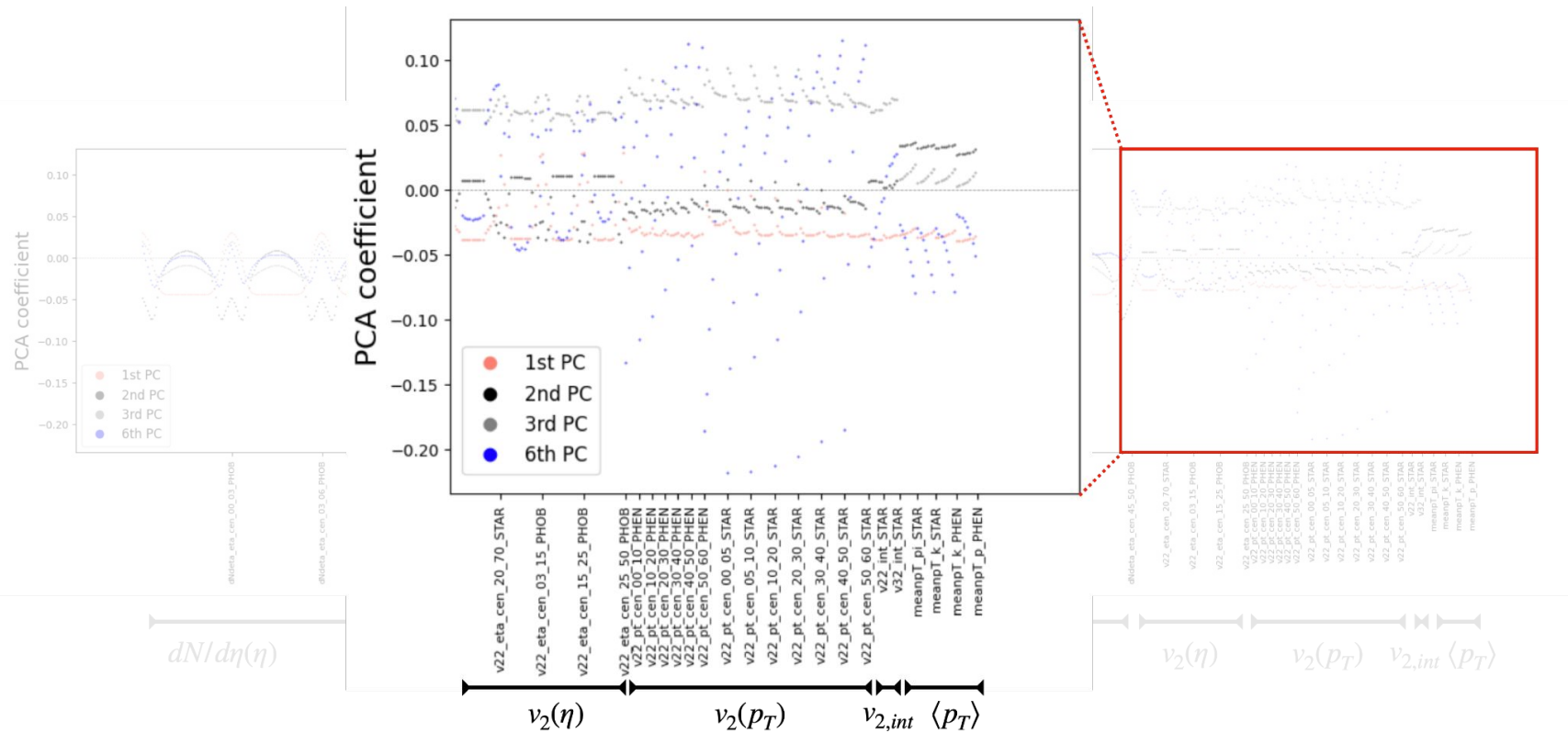
PCA observable coefficients (AuAu)



PCA observable coefficients (AuAu)



PCA observable coefficients (AuAu)



Summary of Experimental Data

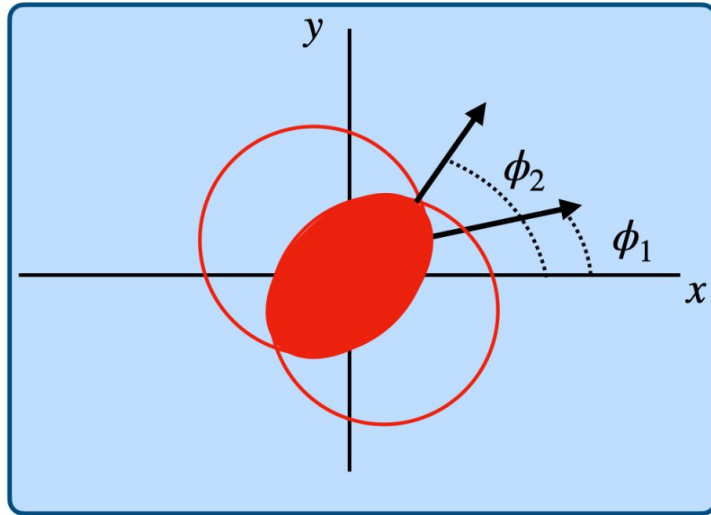
Table II. Listing of all the experimental data used in this work for the default calibration. For each observable we indicate the collision system, the experimental Collaboration, the number of centrality bins used, and the total number of data points used to calculate the likelihood.

observable	system	Experiment	cent. bins	data points
$dN_{ch}/d\eta(\eta)$	Au-Au	PHOBOS [62]	11	594
$v_2(\eta)$	Au-Au	PHOBOS [63]	3	48
$v_2(\eta)$	Au-Au	STAR [64]	1	21
$v_2(p_T)$	Au-Au	PHENIX [65]	6	30
$v_2(p_T)$	Au-Au	STAR [64]	7	56
$v_2(\text{cent.})$	Au-Au	STAR [64]	6	6
$v_3(\text{cent.})$	Au-Au	STAR [66]	6	6
$\pi \langle p_T \rangle (\text{cent.})$	Au-Au	STAR [67]	7	7
$k \langle p_T \rangle (\text{cent.})$	Au-Au	STAR [67]	7	7
$k \langle p_T \rangle (\text{cent.})$	Au-Au	PHENIX [68]	8	8
$p \langle p_T \rangle (\text{cent.})$	Au-Au	PHENIX [68]	8	8
$dN_{ch}/d\eta(\eta)$	d-Au	PHENIX [69]	3	102
$v_2(\eta)$	d-Au	PHENIX [69]	1	14
$v_2(p_T)$	d-Au	PHENIX [30]	1	5
$v_2(p_T)$	d-Au	STAR [70]	1	5
Total				917

Table III. Listing of the experimental data *not* used in calibration but nevertheless computed and compared with in this analysis. For each observable we indicate the collision system, the experimental Collaboration, the number of centrality bins used, and the total number of data points used to calculate the likelihood.

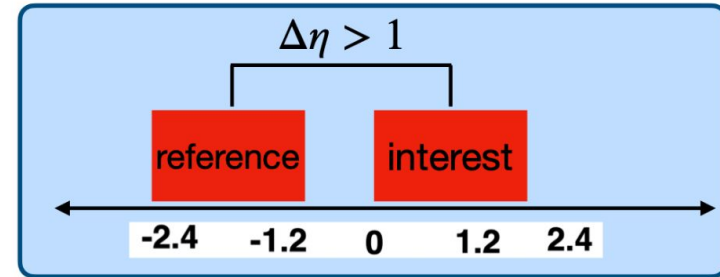
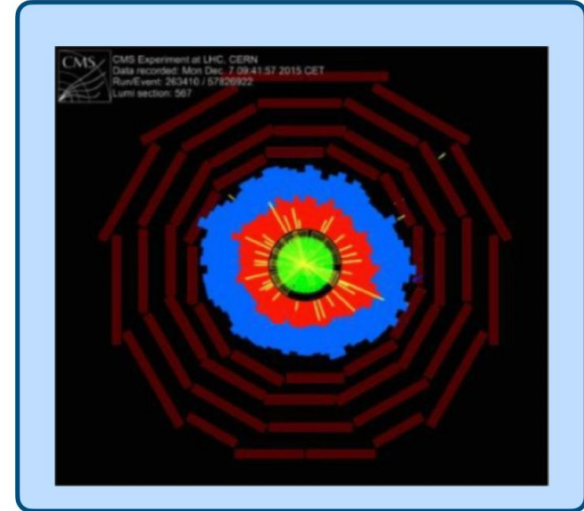
observable	system	Experiment	cent. bins	data points
$dN_{ch}/d\eta(\eta)$	Au-Au	BRAHMS [71]	6	216
$v_3(p_T)$	Au-Au	PHENIX [65]	6	30
$v_4(p_T)$	Au-Au	PHENIX [65]	5	25
$p \langle p_T \rangle (\text{cent.})$	Au-Au	STAR [67]	7	7
$\pi \langle p_T \rangle (\text{cent.})$	Au-Au	PHENIX [68]	8	8
$dE_T/d\eta(\text{cent.})$	Au-Au	PHENIX [72]	12	12
$dN_{ch}/d\eta(\eta)$	$^3\text{He-Au}$	PHENIX [69]	1	26
$v_2(\eta)$	$^3\text{He-Au}$	PHENIX [69]	1	24
$v_2(p_T)$	$^3\text{He-Au}$	STAR [70]	1	7
$v_2(p_T)$	$^3\text{He-Au}$	PHENIX [30]	1	8
$v_3(p_T)$	$^3\text{He-Au}$	STAR [70]	1	7
$v_3(p_T)$	$^3\text{He-Au}$	PHENIX [30]	1	8
$dN_{ch}/d\eta(\eta)$	d-Au	PHOBOS [73]	1	54
$dE_T/d\eta(\text{cent.})$	d-Au	PHENIX [72]	3	3
$v_3(p_T)$	d-Au	PHENIX [30]	1	5
$v_3(p_T)$	d-Au	STAR [70]	1	5
$r_2(p_T)$	d-Au	STAR [32]	1	5
$r_3(p_T)$	d-Au	STAR [32]	1	5
$dN_{ch}/d\eta(\eta)$	p-Au	PHENIX [69]	1	26
$v_2(\eta)$	p-Au	PHENIX [69]	1	24
$v_2(p_T)$	p-Au	STAR [70]	1	7
$v_2(p_T)$	p-Au	PHENIX [30]	1	8
$v_3(p_T)$	p-Au	STAR [70]	1	7
$v_3(p_T)$	p-Au	PHENIX [30]	1	8
Total				535

Experimental Data: a Note on Azimuthal Anisotropy

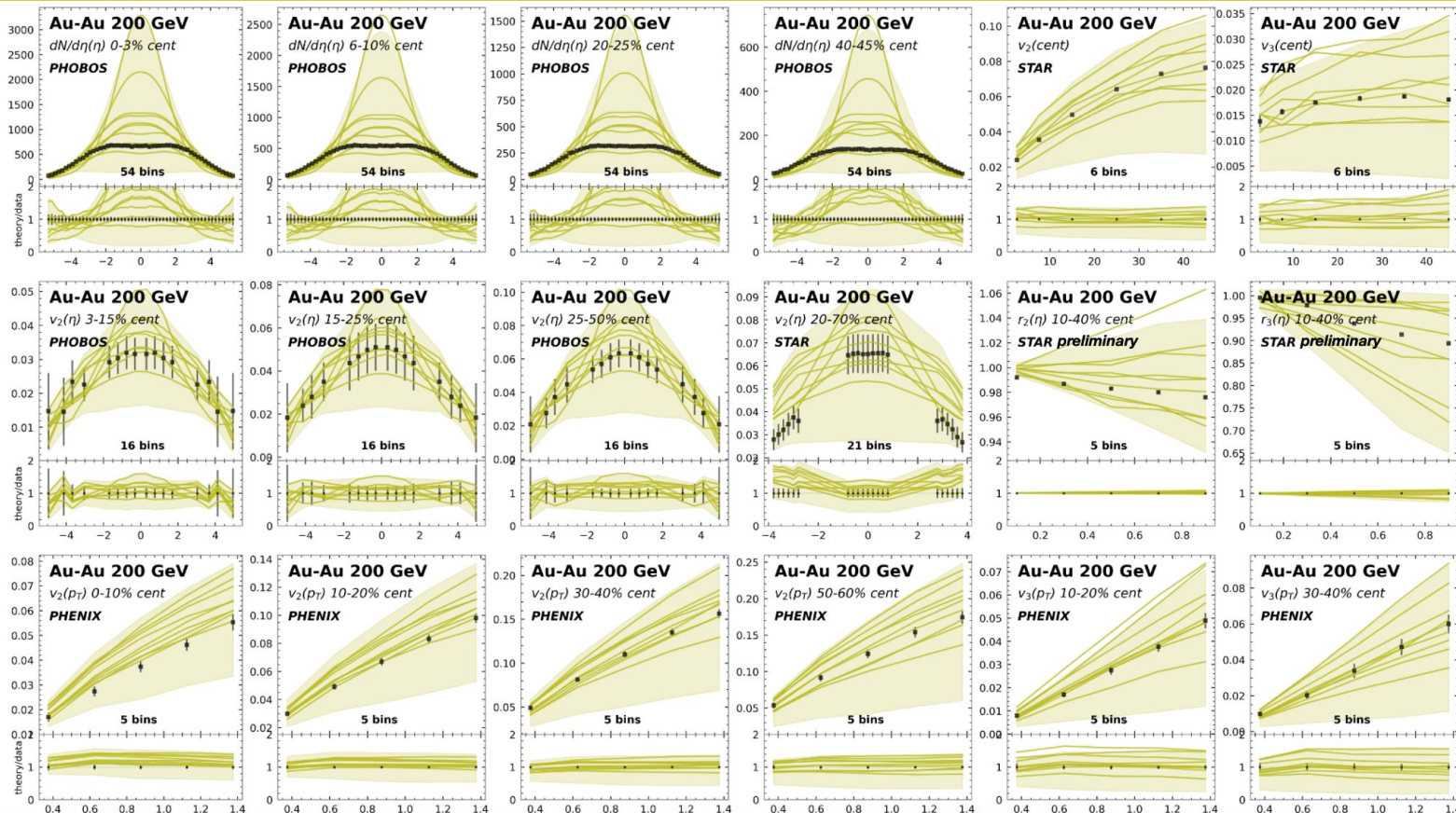


Two-particle spectrum

$$\frac{dN_{pair}}{d\Delta\phi} \propto 1 + 2 \sum V_n \cos(n\Delta\phi)$$

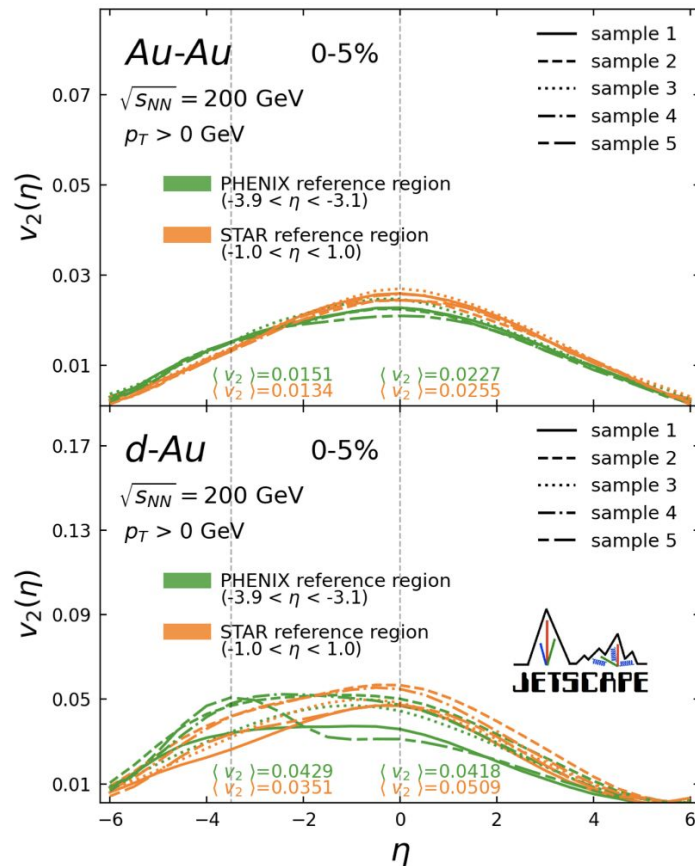


Observable Prior



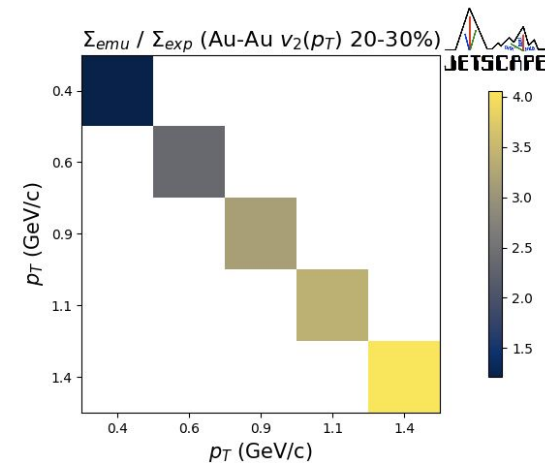
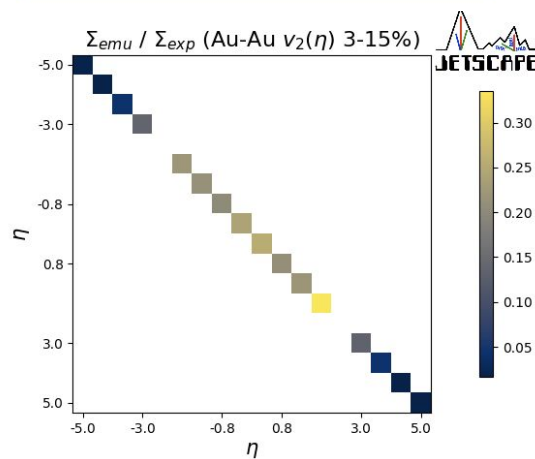
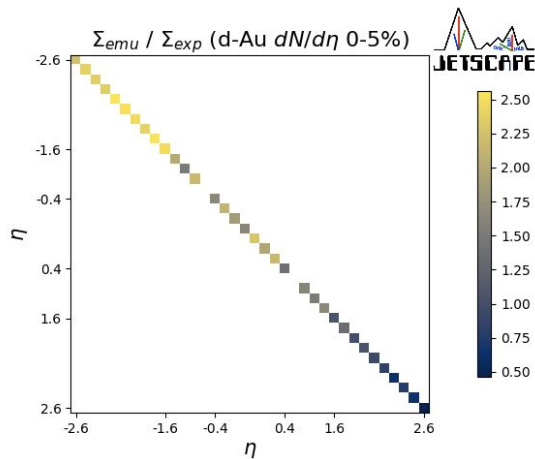
Predictions with the calibrated model

- The effect of selection of the rapidity reference region on $v_2(\eta)$ is $\sim 15\%$ in both Au-Au and d-Au



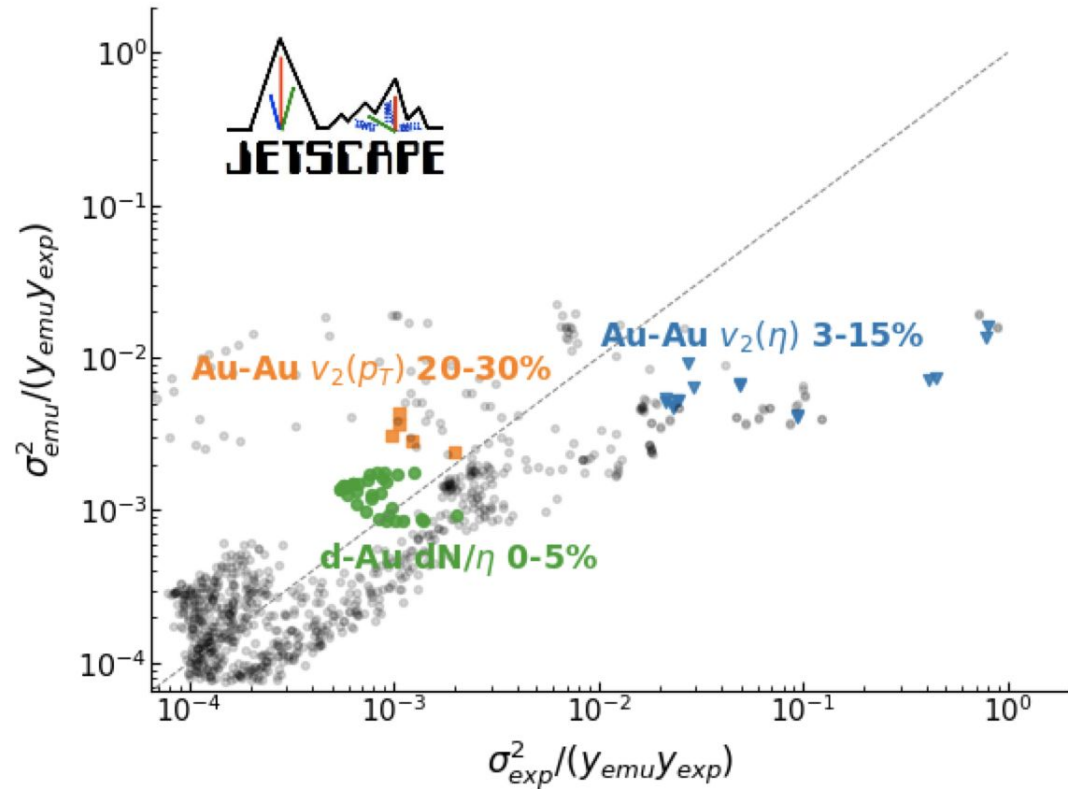
Relative Contribution to Covariance Matrix from Theory and Experiment

$$\Sigma_{emu}/\Sigma_{exp}$$

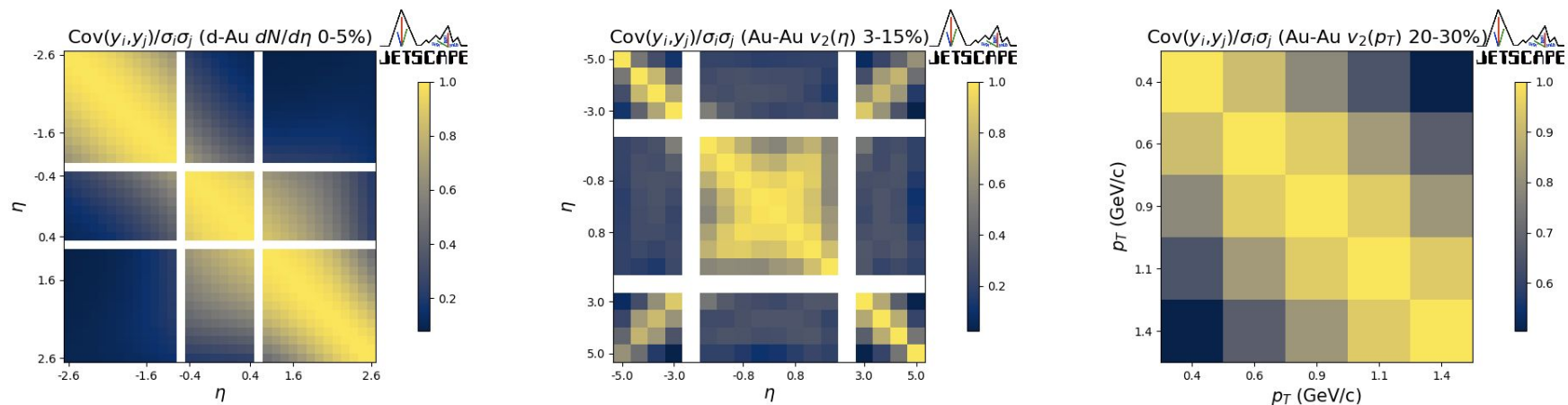


Emulator vs Experimental Variance for all Observables

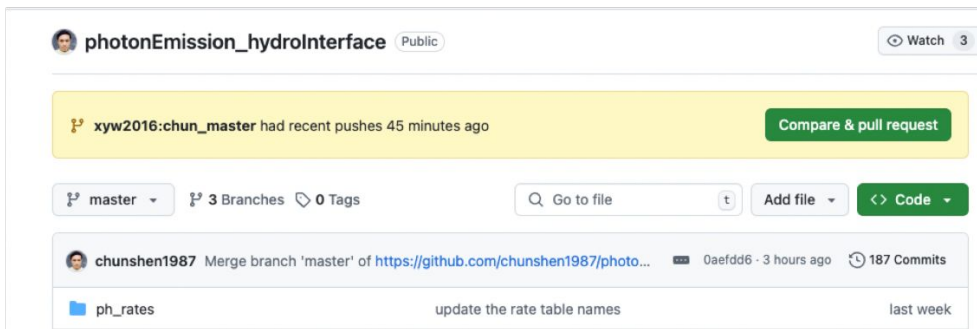
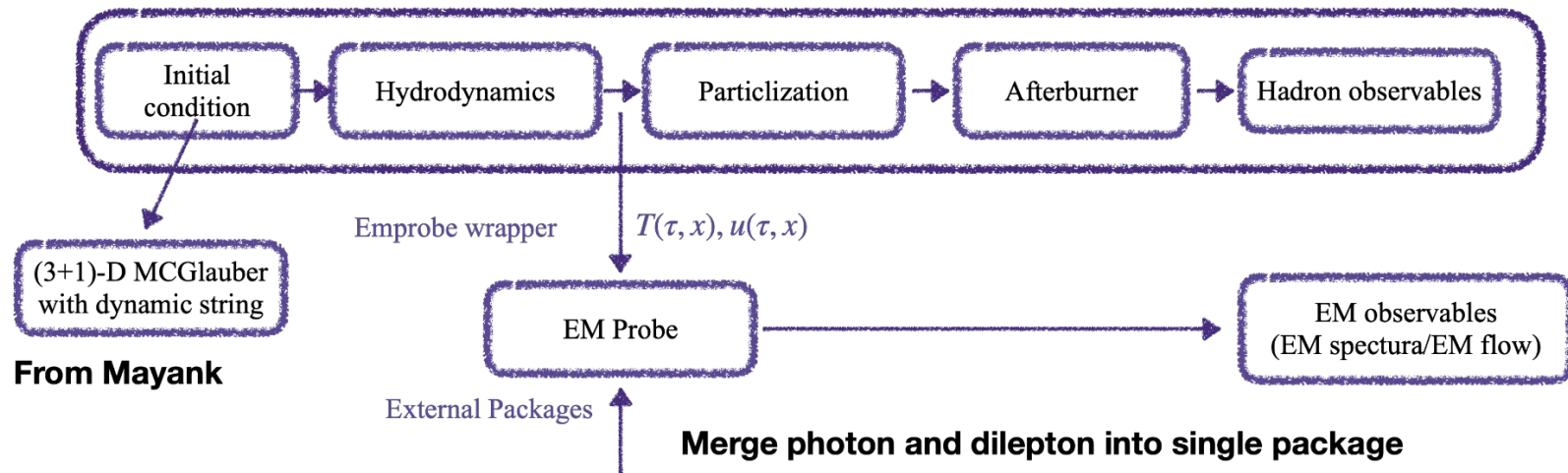
$$\sigma_j^2 / y_j^{emu} y_j^{exp}$$



Correlations between Observable Bins from the Emulator

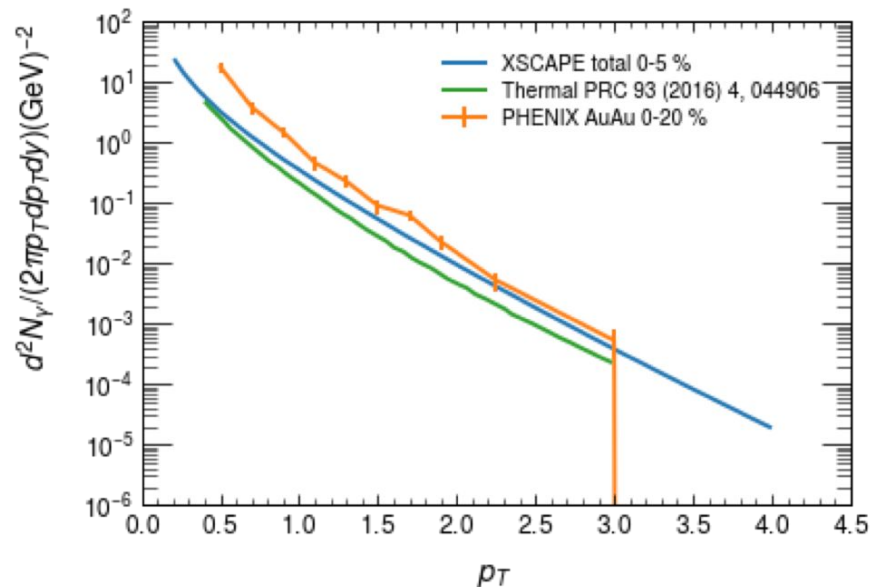


The Improvement of EM Probes in JETSCAPE



Photon and dilepton branch.
It is based on the main branch of XScape-COMP

Photon and dilepton test



Collision system: Au+Au @ 200 GeV

Centrality bins: (0-5%)

Events: one 3+1D hydro event

T_high: 0.165 GeV

T_low: 0.1649 GeV

